

Notes: Parkhomenko (RES, 2026)

Homeownership, Polarization, and Inequality

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1 Big picture

- **Question.** Why have large U.S. cities experienced more job polarization and faster growth in wage inequality than smaller places since 1980?
- **New mechanism.** The paper argues that large-city housing markets help generate the “missing middle.” When house prices grow faster than wages and rents in large cities, middle-income households increasingly cannot buy a house there. Many choose smaller, cheaper commuting zones where ownership is feasible.
- **Core sorting logic.** Low-income households are often unable to buy even in cheap places, while high-income households can buy even in expensive places. Middle-income households are the marginal group whose location choice is most affected by ownership affordability.
- **Empirical facts.** Between 1980 and 2019, commuting zones with faster growth in house prices, price–wage ratios, and price–rent ratios experienced larger declines in middle-skilled employment shares and larger increases in wage inequality.
- **Migration evidence.** CPS interstate migration data show that middle-income households are more likely than low- or high-income households to move to cheaper locations for housing-related reasons.
- **Model.** The paper builds a spatial equilibrium model with heterogeneous skills, tenure choice, city choice, housing supply, rents, house prices, and homeownership constraints.
- **Quantitative exercise.** The model is calibrated to two types of locations—large and small commuting zones—in 1980 and 2019. Large cities have lower housing supply elasticity and higher productivity growth, especially at the top of the skill distribution.
- **Main counterfactual results.** Shutting down skill-biased technical change reduces the large-versus-small city polarization gap by 54% and the inequality gap by 73%. Holding price–wage or price–rent ratios at their 1980 levels reduces excess polarization in large cities by 89–96% and excess inequality by 27–40%.
- **Policy implication.** Housing supply policies that make productive large cities affordable can reduce local polarization and inequality while improving spatial allocation. Policies that only subsidize ownership without expanding supply may reduce some local inequality but can lower output.
- **Main takeaway.** The paper does not reject skill-biased technical change. Instead, it argues that housing affordability amplifies skill-biased technical change by changing which skill groups can remain in large productive cities as homeowners.

2 Data and measurement

2.1 Geography and time period

- The empirical geography is the U.S. commuting zone. A commuting zone is a local labour market and is better suited than counties or cities for studying work, housing, and migration patterns.
- The central historical comparison is between 1980 and 2019. The starting point aligns with the rise of U.S. wage inequality and job polarization. The endpoint avoids the COVID-19 period, when remote work and migration patterns changed sharply.
- The paper studies 465 commuting zones. The empirical regressions are first-difference specifications: the object of interest is the change in labour-market composition or inequality between 1980 and 2019.
- City size is measured by initial CZ population in 1980. Initial size matters because the prior urban literature shows that large cities experienced faster growth in inequality and polarization.

Interpretation: The first-difference design removes time-invariant commuting-zone characteristics. Initial population controls for the fact that big cities were already different before the main period of analysis.

2.2 Skill groups and polarization

- The paper classifies occupations into low-, middle-, and high-skilled groups based on the 1980 occupational wage distribution.
- The middle-skilled group is the middle of the occupation distribution. Polarization is measured by the decline in the employment share of this group.
- The occupational classification is kept constant over time. This means a decline in the middle-skilled share reflects changes in employment composition, not a moving definition of what counts as middle-skilled.
- For each commuting zone, the author computes the middle-skilled employment share in 1980 and in 2019, and then uses the change as the dependent variable.

Interpretation: Job polarization is a local hollowing-out measure. If the middle-skilled share falls more in a location, that place becomes more concentrated in low- and high-skilled occupations.

2.3 Inequality measures

- The paper measures local income inequality using annual wages.
- The two main inequality measures are the Gini coefficient and the variance of log annual wages.
- The primary regression table for inequality focuses on the variance of log wages. Robustness checks use the Gini coefficient and hourly income.

Interpretation: The variance of log wages is especially useful in the model because wages are tied to skill and location-specific productivity. A larger variance indicates more spread in local wage outcomes.

2.4 Housing-cost measures

The paper uses three housing-cost objects:

- **House price growth:** the change in the local house price index.
- **Price–wage ratio growth:** house prices relative to wages.
- **Price–rent ratio growth:** house prices relative to rents.

Interpretation: Each measure captures a slightly different affordability channel. Price growth captures the raw asset-price increase. Price–wage growth measures ownership affordability for workers. Price–rent growth measures the relative attractiveness and financial cost of owning versus renting.

2.5 Migration data

- To study migration reasons, the paper uses the Annual Social and Economic Supplement of the Current Population Survey from 2001 to 2019.
- CPS migration data identify interstate migration and report reasons for moving.
- The key outcome is whether a household moved for housing-related reasons.
- The CPS does not identify commuting-zone moves. The migration analysis therefore works at the state level, but focuses on the mechanism: whether middle-income households move toward cheaper places for housing reasons.

Interpretation: The migration evidence is important because the theory is about sorting. It is not enough to show that expensive places become more unequal; the paper also needs evidence that middle-income households respond to relative housing costs by relocating.

3 Empirical specifications and identification

3.1 City size, polarization, inequality, and housing costs

The first empirical exercise verifies that large CZs experienced more polarization and inequality growth, and that they also experienced faster housing-cost growth. The basic regression is a first-difference regression of the form

$$\Delta Y_i = \alpha + \beta \ln N_{i,1980} + u_i, \quad (1)$$

where Y_i is a local outcome such as the middle-skilled employment share, Gini coefficient, variance of log wages, house prices, price–wage ratio, or price–rent ratio.

Interpretation: The coefficient β asks whether initially larger cities changed more between 1980 and 2019. This exercise motivates the central puzzle: large cities both became more polarized and became less affordable for potential homeowners.

3.2 House prices and polarization

The main polarization regression is

$$\Delta n_i^M = \beta_0 + \beta_1 \Delta Q_i + \beta_2 \ln N_{i,1980} + \beta_3 \chi_{i,1980} + \varepsilon_i, \quad (2)$$

where

- Δn_i^M is the change in the middle-skilled employment share in CZ i ;
- Q_i is house prices, the price–wage ratio, or the price–rent ratio;
- $N_{i,1980}$ is initial population;
- $\chi_{i,1980}$ includes controls such as manufacturing share, female employment share, college share, foreign-born share, and state fixed effects.

Interpretation: A negative β_1 means places with faster housing-cost growth lost more of their middle-skilled employment share. The regression tests whether the relationship survives after comparing cities with similar initial size and labour-market composition.

3.3 Instrumental-variable strategy

The paper also uses an instrument based on the interaction between local land unavailability and the national change in the real interest rate.

- The logic is that lower interest rates raise housing demand and prices more in places where land is scarce or supply is constrained.
- The first stage relates the instrument to house price, price–rent, and price–wage growth.
- The second stage uses the predicted housing-cost growth to estimate the relationship with polarization and inequality.

Interpretation: The IV strategy is designed to isolate housing-cost variation that comes from the interaction of a national financial shock and local supply constraints. This helps address reverse causality, where changes in local skill composition could themselves affect prices.

3.4 House prices and inequality

The inequality regression is

$$\Delta I_i = \beta_0 + \beta_1 \Delta Q_i + \beta_2 \ln N_{i,1980} + \beta_3 \chi_{i,1980} + \varepsilon_i, \quad (3)$$

where ΔI_i is the change in the variance of log annual wages.

Interpretation: If higher housing costs push out middle-income households, the remaining local wage distribution should become more dispersed. Thus the same housing-cost shock should be associated with both lower middle-skilled share and higher wage inequality.

3.5 Migration regression

The CPS migration analysis estimates a logit model in which the outcome is whether household h moves from state i to state j for housing-related reasons. The key regressors are income-quintile indicators interacted with relative housing-cost ratios between origin and destination:

$$M_{hij,t+1} = \Lambda \left(\sum_{q=1}^5 \delta_1^q I_{h,it}^q + \sum_{q=1}^5 \delta_2^q \ln \left(\frac{Q_{it}}{Q_{jt}} \right) I_{h,it}^q + \delta_3 \ln(1 + D_{ij}) + \delta_4 X_{hij,t+1} + \phi_i + \phi_j + \phi_{t+1} \right). \quad (4)$$

Interpretation: The coefficient of interest is the interaction between income rank and relative housing costs. The theory predicts the largest effect for middle-income households, because they are most likely to be priced out of ownership in expensive places but able to own in cheaper ones.

3.6 Evidence credibility

- The reduced-form evidence is not used to estimate the quantitative model, so it provides an independent empirical motivation.
- The OLS results show a strong conditional relationship between local affordability and local polarization/inequality.
- The IV results suggest that part of this relationship can be interpreted as causal, especially for prices and price-wage ratios.
- The migration evidence shows the micro-level channel: middle-income households are differentially likely to move for housing-related reasons when destination states are cheaper.

Interpretation: The evidence does not claim that housing prices are the only cause of polarization. Instead, it establishes a consistent set of facts that a model of homeownership, location choice, and skill heterogeneity must explain.

4 Theoretical model

4.1 Households and preferences

Households live for T years, draw skill $s \in (0, 1]$, draw location-preference shocks, and choose where to live and whether to own or rent. The common component of utility in city i is

$$v_i(c, h) = \left(\frac{c}{1-\gamma} \right)^{1-\gamma} \left(\frac{h}{\gamma} \right)^{\gamma} X_i, \quad (5)$$

where c is consumption of the tradable good, h is housing consumption, X_i is local amenities, and γ is the housing expenditure share.

Interpretation: Amenities enter multiplicatively, so a city can attract households through higher wages, lower housing costs, or higher amenities. Housing enters directly in utility, which makes tenure and price differences central.

4.2 Renters

A renter with skill s in city i solves

$$\max_{c, h} v_i(c, h) \quad \text{s.t.} \quad w_i(s) = c + r_i h. \quad (6)$$

The housing demand of renters is

$$h_i^R(s) = \gamma \frac{w_i(s)}{r_i}, \quad (7)$$

and indirect utility is

$$v_i^R(w_i(s), r_i) = \frac{w_i(s) X_i}{r_i^{\gamma}}. \quad (8)$$

Interpretation: Renting has no minimum-size or payment-to-income constraint. This makes renting feasible for low-income households who cannot afford the minimum owned unit.

4.3 Owners and ownership constraints

Homeowners buy a dwelling at price p_i , sell it at the end of life, and pay a city-specific user cost δ_i . The budget constraint is

$$w_i(s) = c + \delta_i p_i h. \quad (9)$$

Ownership is subject to two constraints:

$$h \geq \bar{h} \quad \text{minimum owner-occupied size,} \quad (10)$$

$$p_i h \leq \lambda w_i(s) \quad \text{payment-to-income constraint.} \quad (11)$$

Interpretation: These constraints generate a threshold structure. Low-skill households cannot buy at all. Middle-skill households may buy only the minimum house. High-skill households can buy their unconstrained optimal house.

4.4 Tenure thresholds

The paper's Lemmas 1 and 2 show that, under conditions on the price-rent ratio, there is a unique ownership threshold s_i^* and a full-ownership threshold s_i^{**} :

- workers with $s < s_i^*$ rent;
- workers with $s \in [s_i^*, s_i^{**})$ are marginal owners who buy the minimum size $\bar{h}$;
- workers with $s \geq s_i^{**}$ are full owners who buy more than $\bar{h}$.

Interpretation: The key comparative static is that s_i^* and s_i^{**} are higher in places where housing is less affordable. A higher price-wage ratio makes it harder to buy; a higher price-rent ratio makes owning less attractive relative to renting.

4.5 Location choice

Household n chooses the city that maximizes idiosyncratic-preference-weighted utility:

$$\max_i \{ \xi_{ni} v_i(w_i(s), p_i, r_i) \}. \quad (12)$$

Because ξ_{ni} follows a Fréchet distribution, the probability that a skill- s worker chooses city i is

$$\pi_i(s) = \frac{v_i(w_i(s), p_i, r_i)^\epsilon}{\sum_{j \in I} v_j(w_j(s), p_j, r_j)^\epsilon}. \quad (13)$$

The supply of skill- s workers in city i is

$$N_i(s) = \pi_i(s) \phi(s). \quad (14)$$

Interpretation: A city attracts workers when it offers high wages, high amenities, low rents, low house prices, or a more valuable tenure option. The important feature is that location probabilities vary by skill because tenure feasibility varies by skill.

4.6 Location-tenure interaction

Suppose city i is expensive and city j is cheap, so the ownership threshold is higher in city i . Workers with $s \in [s_j^*, s_i^*)$ can buy in city j but must rent in city i . Their relative location probability includes a direct comparison of renting in one city versus owning in the other:

$$\frac{\pi_i(s)}{\pi_j(s)} = \left[\frac{v_i^R(w_i(s), r_i)}{v_j^O(w_j(s), p_j)} \right]^\epsilon. \quad (15)$$

Interpretation: This equation is the heart of the mechanism. Middle-skill workers compare not only wages and amenities but also tenure status. If moving lets them become homeowners, the cheaper city becomes more attractive.

4.7 Production and wages

Final goods are produced with workers of different skills as perfect substitutes:

$$Y_i = A_i \int_0^1 a_i(s) N_i(s) ds. \quad (16)$$

Local productivity includes agglomeration:

$$A_i = \bar{A}_i N_i^\theta. \quad (17)$$

Perfect competition implies

$$w_i(s) = A_i a_i(s). \quad (18)$$

Interpretation: The model can include skill-biased technical change through changes in $a_i(s)$ or the dispersion of returns to skill. But the paper emphasizes that tenure choice can generate local polarization even without relying entirely on production complementarities.

4.8 Housing supply and price–rent ratios

Developers produce housing using land and the final good:

$$H_i = (\phi_i L_i)^{\eta_i} K_i^{1-\eta_i}. \quad (19)$$

The price–rent ratio comes from real estate managers’ no-arbitrage condition:

$$\frac{p_{i,t}}{r_{i,t}} = \sum_{\tau=t}^{\infty} [\beta(1 - \tilde{\delta}_i/T)]^{\tau-t} = \frac{1}{1 - \beta(1 - \tilde{\delta}_i/T)}. \quad (20)$$

Interpretation: Lower user costs raise the price–rent ratio. Housing supply in large cities is less elastic because the land share is higher, so demand shocks produce larger price changes.

4.9 Equilibrium

A stationary spatial equilibrium consists of skill-specific local labour supply $N_i(s)$, rents r_i , and prices p_i such that:

- workers choose locations according to the Fréchet location-choice rule;
- prices and rents satisfy the real estate managers’ no-arbitrage condition;
- local housing markets clear, with demand from renters, marginal owners, and full owners.

Interpretation: The equilibrium simultaneously determines who lives where, who rents, who owns, local wages, local rents, and local prices.

4.10 Main theoretical result

The paper defines local low-, middle-, and high-skilled shares as fractions of the skill distribution below s' , between s' and s'' , and above s'' :

$$n_i^L(s') = \frac{1}{N_i} \int_0^{s'} N_i(s) ds, \quad (21)$$

$$n_i^H(s'') = \frac{1}{N_i} \int_{s''}^1 N_i(s) ds, \quad (22)$$

$$n_i^M(s', s'') = 1 - n_i^L(s') - n_i^H(s''). \quad (23)$$

Proposition 1 shows that a city with higher price–wage and price–rent ratios has a higher low-skilled share and, under an additional bounded price–rent condition, a higher high-skilled share. Therefore, it has a lower middle-skilled share and greater polarization.

Interpretation: The result does not require skill complementarities or task-based technological change. It comes from the fact that the middle of the skill distribution has the strongest tenure-driven incentive to avoid expensive places.

5 Quantitative model and calibration

5.1 Two-location quantitative economy

The quantitative model represents the U.S. economy with two location groups:

- large commuting zones;
- small commuting zones.

The model is calibrated separately for 1980 and 2019. Each period is interpreted as a stationary equilibrium.

Interpretation: The two-location structure is simple but targeted. The goal is not to match every city but to explain why large and small CZs evolved differently.

5.2 Skill distribution and skill-biased technical change

- The skill distribution is calibrated so that the model can match the low-, middle-, and high-skilled employment shares in each location group and year.
- Local returns to skill can differ across large and small CZs and over time.
- Skill-biased technical change is represented by growth in parameters that raise returns to high skills, especially in large cities.

Interpretation: In the benchmark model, SBTC remains an important force. The paper’s contribution is to show that housing affordability magnifies its spatial consequences.

5.3 Housing supply calibration

- The land-share parameter η_i differs between small and large CZs.
- Large CZs have higher land shares and therefore less elastic housing supply.
- Developer productivity and land availability are calibrated jointly because the model only needs their product.
- Price–wage ratios are used to discipline the housing supply side.

Interpretation: A productivity or skill-demand shock has a larger effect on prices in large cities because housing supply is less elastic. This is why the same labour-market forces can produce different affordability outcomes across city sizes.

5.4 Owner and rental user costs

- Owners’ expenditures δ_i represent the user cost of owner-occupied housing.
- Real estate managers’ expenditures $\tilde{\delta}_i$ determine price–rent ratios.

- The model allows these costs to vary by location group and year to match observed price–rent and price–wage ratios.

Interpretation: Separating owner costs and rental-manager costs allows the model to distinguish price–wage affordability from price–rent valuation effects.

5.5 Model fit

The model is disciplined to match:

- middle-skilled employment shares in large and small CZs;
- variances of log wages in large and small CZs;
- price–rent ratios;
- price–wage ratios;
- homeownership rates and other aggregate moments.

Interpretation: Figure 4 demonstrates the key calibration targets. The benchmark model reproduces the widening large-small gaps in polarization, inequality, and affordability.

6 Counterfactual experiments

6.1 No skill-biased technical change

The first counterfactual fixes the local returns-to-skill parameters at their 1980 values and computes the 2019 equilibrium.

Interpretation: This removes the standard production-side explanation. If the large-small gaps shrink, then SBTC is important for explaining the spatial divergence.

6.2 Keeping price–wage ratios at 1980 levels

The second counterfactual keeps price–wage ratios at their 1980 levels by allowing faster housing construction productivity or greater effective land supply.

Interpretation: This is the most direct housing-supply counterfactual. It asks what would happen if large cities had become less expensive relative to wages despite rising productivity.

6.3 Keeping price–rent ratios at 1980 levels

The third counterfactual holds price–rent ratios fixed by adjusting the user cost of housing.

Interpretation: This is closer to changing the financial cost of ownership relative to renting. It does not necessarily increase housing supply, so it can have different aggregate-output effects.

6.4 Keeping both ratios fixed

The fourth counterfactual holds both price–wage and price–rent ratios at their 1980 levels.

Interpretation: This combines supply-side affordability with ownership-cost changes. It isolates how much of the spatial polarization and inequality gap is amplified by the full deterioration in affordability.

6.5 Counterfactual results

- Removing SBTC reduces the large-small gap in the middle-skilled share decline by 54%.
- Removing SBTC reduces the large-small gap in the variance of log wages by 73%.
- Holding price–wage or price–rent ratios at 1980 levels reduces excess polarization in large cities by 89–96%.
- Holding these ratios fixed reduces excess inequality by 27–40%.
- Middle-skilled workers are the largest welfare beneficiaries because they regain access to homeownership in productive large cities or avoid having to move to less productive cities.

Interpretation: The results say that SBTC is important but not sufficient. Housing-market affordability strongly determines how SBTC translates into the spatial distribution of skills.

6.6 Robustness and sensitivity

The paper discusses multiple robustness exercises:

- allowing multiple locations within a city;
- changing the labour supply elasticity;
- changing the payment-to-income constraint;
- removing agglomeration externalities;
- fixing housing expenditure shares;
- fixing the minimum owner-occupied size.

Interpretation: The main qualitative result survives: when ownership affordability deteriorates in large cities, middle-skilled households are pushed out more than other groups, amplifying polarization and inequality.

7 Tables, figures, and original excerpts

7.1 Table 1 + Table 3: city size facts and first-stage variation

Read:

- Table 1 shows that larger CZs had larger declines in the middle-skilled share and larger increases in inequality from 1980 to 2019.
- The same table shows that larger CZs had faster growth in house prices, price–wage ratios, and price–rent ratios.
- Table 3 shows a strong first stage: the land-unavailability-by-interest-rate instrument predicts price growth, price–rent growth, and price–wage growth.

- The first-stage relationships are statistically strong, with high F-statistics in the later IV tables.

Economic message: The empirical starting point is a joint spatial pattern: big cities became more polarized, more unequal, and less affordable. The IV first stage supplies quasi-exogenous affordability variation for the causal exercises.

TABLE 1
Relationships between city size, polarization, inequality, and house prices

	(1) Mid-skl. chg.	(2) Gini coeff. chg.	(3) Var. log w chg.
Panel A: Polarization and inequality			
Log initial population	-1.899*** (0.133)	1.254*** (0.0765)	2.745*** (0.182)
R-squared	0.298	0.363	0.405
	(1) Log price chg.	(2) Log p/w chg.	(3) Log p/r chg.
Panel B: Housing costs			
Log initial population	0.0339*** (0.0114)	0.0311*** (0.00916)	0.101*** (0.00955)
R-squared	0.0310	0.0380	0.187

The table shows the results from first-difference OLS regressions for the 1980–2019 period. In panel A, Column (1) reports the coefficient from the regression of the change in 100× the middle-skilled share on log CZ population in 1980. Columns (2) and (3) report the coefficients for the change in 100× the Gini coefficient of annual wages and 100× the variance of log annual wages, respectively. In panel B, Columns (1)–(3) report the coefficients for the change in log house prices, price–wage ratios, and price–rent ratios. The number of observations in each regression is 465 (the number of CZs). Robust standard errors are reported in parentheses. *, **, and *** indicate 10, 5, and 1% significance levels.

(a) Table 1: city size, polarization, inequality, and housing prices.

TABLE 3
First-stage regressions

	(1) Price chg.	(2) P/r chg.	(3) P/w chg.
Land and interest rate IV	4.087*** (0.558)	0.435*** (0.0912)	1.223*** (0.145)
R-squared	0.132	0.0550	0.195

Note: The table shows the results from first-difference regressions for the 1980–2019 period. Columns (1), (2), and (3) report the coefficients from regressions of the change in housing prices, price–rent indices, and price–wage indices, respectively, on the instrumental variable that interacts the Lutz and Sand (2023) land unavailability index with the change in real interest rate. The number of observations in each regression is 465 (the number of CZs). Robust standard errors are reported in parentheses. *, **, and *** indicate 10, 5, and 1% significance levels.

(b) Table 3: first-stage regressions.

7.2 Table 2 + Table 4: housing costs, polarization, and inequality

Read:

- Table 2 shows that house price growth is negatively related to the change in the middle-skilled share. Faster price growth means larger polarization.
- The same negative relation appears for price–rent and price–wage ratios.
- The 2SLS coefficients are generally larger than OLS coefficients, consistent with attenuation or reverse causality in OLS.
- Table 4 shows the parallel inequality result: faster housing-cost growth is associated with larger increases in the variance of log wages.
- The price–wage and price results remain important after initial population and additional controls are included.

Economic message: These tables are the core reduced-form evidence. Housing affordability is linked to both pieces of the paper’s title: polarization and inequality.

7.3 Figure 1: migration for housing-related reasons

Read:

- Panel A uses the full CPS sample and shows marginal effects by income quintile.
- Panel B focuses on interstate migrants and shows similar patterns conditional on moving.
- The second through fourth income quintiles show the largest response to relative housing costs.
- The fifth quintile is less sensitive because high-income households can still buy in expensive places.
- The first quintile is less sensitive because low-income households often cannot buy even in cheap places.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Prices						
Price change	-1.002*** (0.183)	-0.585*** (0.165)	-1.006*** (0.197)	-2.899*** (0.558)	-2.387*** (0.524)	-4.052*** (1.201)
Log initial population		-1.784*** (0.135)	-1.275*** (0.161)		-1.430*** (0.208)	-1.596*** (0.222)
Mean of dependent variable	-2.860	-2.860	-2.860	-2.860	-2.860	-2.860
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.0700	0.320	0.595			
First-stage F-statistic				53.66	52.61	17.98
Panel B: Price-rent ratios						
Price-rent ratio change	-11.26*** (1.110)	-7.127*** (1.181)	-3.135*** (1.426)	-27.25*** (5.375)	-26.89*** (6.862)	-47.84*** (22.45)
Log initial population		-1.424*** (0.141)	-1.077*** (0.177)		-0.107 (0.520)	0.232 (0.745)
Mean of dependent variable	-2.860	-2.860	-2.860	-2.860	-2.860	-2.860
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.240	0.375	0.574			
First-stage F-statistic				22.74	15.53	4.319
Panel C: Price-wage ratios						
Price-wage ratio change	-5.335*** (0.656)	-3.832*** (0.652)	-4.458*** (0.845)	-9.688*** (1.656)	-7.830*** (1.486)	-14.06*** (3.974)
Log initial population		-1.729*** (0.134)	-1.201*** (0.160)		-1.552*** (0.173)	-1.271*** (0.176)
Mean of dependent variable	-2.860	-2.860	-2.860	-2.860	-2.860	-2.860
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.120	0.357	0.600			
First-stage F-statistic				73.42	68.89	21.85

Notes: The table shows the results from first-difference regressions for the 1980–2019 period. Panel A shows results for the house price index, panel B shows results for price-rent ratios, and panel C shows results for price-wage ratios. Column (1) reports the coefficients from the OLS regression of the change in 100× the middle-skilled share on the change in prices. Column (2) includes initial CZ population as a control. Column (3) adds manufacturing share, female share, college share, foreign-born share, and state dummy as additional controls. Columns (4)–(6) report the results from 2SLS estimation. The number of observations in each regression is 465 (the number of CZs). Robust standard errors are reported in parentheses. *, **, and *** indicate 10, 5, and 1% significance levels.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Prices						
Price change	2.290*** (0.269)	1.728*** (0.204)	1.709*** (0.230)	2.350*** (0.524)	1.470*** (0.450)	2.677** (1.090)
Log initial population		2.405*** (0.146)	1.711*** (0.203)		2.456*** (0.166)	1.813*** (0.229)
Mean of dependent variable	10.02	10.02	10.02	10.02	10.02	10.02
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.237	0.534	0.666			
First-stage F-statistic				53.66	52.61	17.98
Panel B: Price-rent ratios						
Price-rent ratio change	11.72*** (1.641)	4.641*** (1.360)	5.816*** (1.637)	22.09*** (5.427)	16.56*** (5.963)	31.61* (17.30)
Log initial population		2.436*** (0.169)	1.361*** (0.225)		1.642*** (0.425)	0.605 (0.576)
Mean of dependent variable	10.02	10.02	10.02	10.02	10.02	10.02
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.169	0.426	0.628			
First-stage F-statistic				22.74	15.53	4.319
Panel C: Price-wage ratios						
Price-wage ratio change	7.735*** (1.241)	5.563*** (0.864)	4.314*** (1.113)	7.853*** (1.867)	4.822*** (1.535)	9.292** (4.027)
Log initial population		2.499*** (0.154)	1.562*** (0.219)		2.531*** (0.164)	1.599*** (0.220)
Mean of dependent variable	10.02	10.02	10.02	10.02	10.02	10.02
Model	OLS	OLS	OLS	2SLS	2SLS	2SLS
Additional controls	No	No	Yes	No	No	Yes
R-squared	0.164	0.487	0.634			
First-stage F-statistic				73.42	68.89	21.85

Notes: The table shows the results from first-difference regressions for the 1980–2019 period. Panel A shows results for the house price index, panel B shows results for price-rent ratios, and panel C shows results for price-wage ratios. Column (1) reports the results from the OLS regression of the change in 100× the variance of log annual wages on the change in prices. Column (2) includes initial CZ population as a control. Column (3) adds manufacturing share, female share, college share, foreign-born share, and state dummy as additional controls. Columns (4)–(6) report the results from 2SLS estimation. The number of observations in each regression is 465 (the number of CZs). Robust standard errors are reported in parentheses. *, **, and *** indicate 10, 5, and 1% significance levels.

Economic message: Migration evidence validates the mechanism. The marginal household whose location is distorted by homeownership affordability is in the middle, not at the bottom or top.

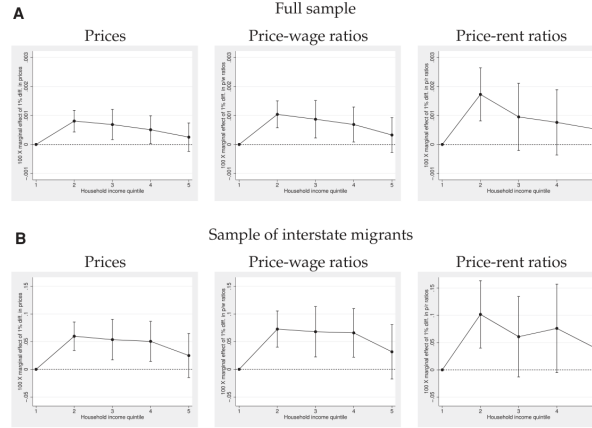


FIGURE 1
Marginal effects on migration for housing-related reasons

Note: The figure reports marginal effects from coefficients β_1^i from regression (3) on the full sample of 1,265,832 observations (panel A) or the sample of 15,815 interstate migrants (panel B). The left plot shows 100× marginal effects on the probability of moving for housing-related reasons of the log ratio of house prices in the location of origin to the prices in the location of destination for each quintile in the household income distribution at the location of origin. The center plot shows the marginal effects of price-wage ratios, and the right plot shows the marginal effects of price-rent ratios. Vertical bars represent the 95% confidence interval. Standard errors of marginal effects are computed using the Delta method. Standard errors of the underlying logit regression are clustered by the state of origin.

Figure 1: marginal effects on migration for housing-related reasons.

7.4 Original model equations: household problem and tenure thresholds

Read:

- The renter problem is simple and frictionless.
- The owner problem adds minimum-size and payment-to-income constraints.
- The owner utility and housing demand functions become piecewise because different constraints bind for

different skill levels.

- Lemma 2 produces two endogenous skill thresholds: marginal ownership and full ownership.

Economic message: The model turns housing affordability into a threshold object. Changes in prices and rents shift thresholds and therefore move whole regions of the skill distribution across cities.

is the common component of the utility function and $0 < \gamma < 1$.

Households make three choices: location, housing tenure, and consumption of goods and housing. Location and tenure choices are made at the beginning of life and households cannot change their decision afterwards. Because the economy is assumed to be in a stationary equilibrium, because workers receive all shocks (skill and location preference) once, because they make location and tenure choices at the beginning of life, and because they live hand-to-mouth, their problem is essentially static. Therefore, unless noted otherwise, all flow variables and parameters are meant to represent present discounted values over T years and are written without a time subscript, whereas variables that can change at annual frequency have a t -subscript.

3.1.1. Consumption of goods and housing. I now solve the household utility maximization problem, conditional on the tenure choice.

Renters. A household that chooses to rent solves

$$\max_{c, h} v_i(c, h) \quad \text{subject to: } w_i(s) = c + r_i h, \quad (5)$$

where $w_i(s)$ is the wage of a worker with skill s in city i and r_i is the rent per square foot in city i . The housing demand function is $h = \gamma w_i(s)/r_i$ and the indirect utility function of a renter is

$$v_i^R(w_i(s), r_i) = \frac{w_i(s) X_i}{r_i^\gamma}. \quad (6)$$

Homeowners. A household that chooses to own buys a house at the beginning of their life at price p_i and sells it at the end of their life. Since the equilibrium is stationary, they sell at the same price p_i . Owning a house has a city-specific user cost δ_i that represents depreciation, property taxes, mortgage interest payments, etc. Thus, the budget constraint of a homeowner is $w_i(s) + p_i h = c + (1 + \delta_i) p_i h$ which can be simplified as $w_i(s) = c + \delta_i p_i h$. Besides the budget constraint, homeowners are subject to two additional constraints. First, owner-occupied houses cannot be smaller than $\bar{h} > 0$. This is the *minimum-size constraint*. Second, a household cannot spend more than a fraction $\lambda < 1$ of their labour earnings on purchasing a house. This is the *payment-to-income (PTI) constraint*. Thus, owners solve

$$\max_{c, h} v_i(c, h) \quad \text{subject to: } w_i(s) = c + \delta_i p_i h, \quad (7)$$

(a) Preferences, renter problem, and owner constraints.

The solution to the homeowner's problem yields the following housing consumption function:

$$h = \begin{cases} \frac{\gamma}{\delta_i} \frac{w_i(s)}{p_i} & \text{if } \lambda > \frac{\gamma}{\delta_i} \text{ and } w_i(s) \geq \frac{\delta_i p_i \bar{h}}{\gamma}, \\ \bar{h} & \text{if } \lambda > \frac{\gamma}{\delta_i} \text{ and } \delta_i p_i \bar{h} < w_i(s) < \frac{\delta_i p_i \bar{h}}{\gamma}, \\ \lambda \frac{w_i(s)}{p_i} & \text{if } \lambda \leq \frac{\gamma}{\delta_i} \text{ and } w_i(s) \geq \frac{p_i \bar{h}}{\lambda}, \\ 0 & \text{otherwise.} \end{cases} \quad (10)$$

This function describes four cases. First, a homeowner's wage may be high enough to spend just a fraction γ/δ_i of income on housing and be able to afford a house at least as large as $\bar{h}$. Second, the wage may be relatively low and buying even the minimum-size house requires spending a fraction of income greater than γ/δ_i . These two cases occur when the PTI constraint is sufficiently lax, $\lambda > \gamma/\delta_i$. Third, if $\lambda \leq \gamma/\delta_i$, households spend a fraction λ of income on housing.¹⁹ In the fourth case, the wage is so low that even if the household spends all of their income on housing, they would not be able to afford the minimum-size property. Their housing consumption is zero, which violates the positive consumption requirement. These four cases result in the following indirect utility function:

$$v_i^O(w_i(s), p_i) = \begin{cases} \frac{w_i(s)}{\delta_i p_i^\gamma} \left(\frac{\delta_i - \gamma}{1 - \gamma} \right)^{1-\gamma} X_i & \text{if } \lambda > \frac{\gamma}{\delta_i} \text{ and } w_i(s) \geq \frac{\delta_i p_i \bar{h}}{\gamma}, \\ \left(\frac{w_i(s) - \delta_i p_i \bar{h}}{1 - \gamma} \right)^{1-\gamma} \left(\frac{\bar{h}}{\gamma} \right)^\gamma X_i & \text{if } \lambda > \frac{\gamma}{\delta_i} \text{ and } \delta_i p_i \bar{h} < w_i(s) < \frac{\delta_i p_i \bar{h}}{\gamma}, \\ \frac{w_i(s)}{p_i^\gamma} \left(\frac{1 - \delta_i \lambda}{1 - \gamma} \right)^{1-\gamma} \left(\frac{\lambda}{\gamma} \right)^\gamma X_i & \text{if } \lambda \leq \frac{\gamma}{\delta_i} \text{ and } w_i(s) \geq \frac{p_i \bar{h}}{\lambda}, \\ -\infty & \text{otherwise.} \end{cases} \quad (11)$$

3.1.2. Housing tenure choice. Indirect utility functions (6) and (11) result in different utility levels for the same wage, depending on the housing tenure choice. A worker with skill s in city i chooses to rent if it yields higher utility and to own otherwise. Hence, the indirect utility function of such a worker is

$$v_i(w_i(s), p_i, r_i) \equiv \max \{v_i^O(w_i(s), p_i), v_i^R(w_i(s), r_i)\}. \quad (12)$$

The next two results characterize tenure choice. Lemma 1 provides necessary and sufficient conditions for homeownership to be an optimal choice for some households. If any of these

(b) Owner demand and indirect utility.

7.5 Figure 2 + Figure 3: tenure choice and the missing middle

Read:

- Figure 2 shows that renters occupy the low-skill region, marginal owners occupy the middle region, and full owners occupy the high-skill region.
- The ownership threshold s^* determines who can switch from renting to owning.
- Figure 3 illustrates the missing-middle mechanism. Workers who can own in city 2 but not city 1 may relocate to city 2.
- This raises the low- and high-skilled shares in the expensive city and lowers its middle-skilled share.

Economic message: The theory is a sorting model. Expensive cities do not merely become places where everyone pays more; they become places where middle-income potential homeowners are missing.

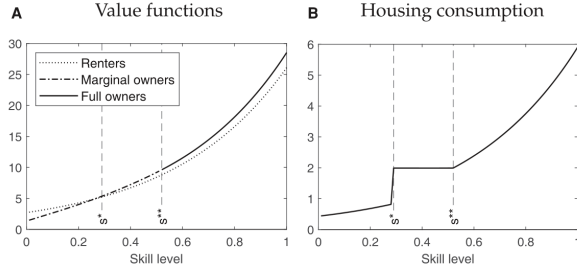


FIGURE 2

Tenure choice and housing consumption

Notes: Panel A shows utility levels for each skill level implied by value functions of renters, marginal owners, and full owners. Note that full ownership is not a feasible choice for $s < s^*$; hence, the value of a full owner is only shown for $s \geq s^*$. Panel B shows housing consumption for each skill level. Both panels display skill thresholds for marginal ownership (s^*) and full ownership (s^{**}).

(a) Figure 2: tenure choice and housing consumption.

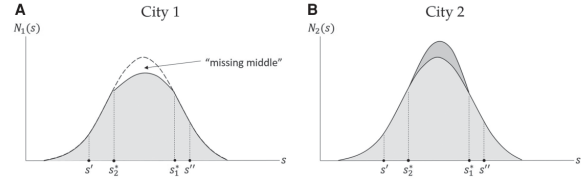


FIGURE 3

Illustration of Proposition 1

Notes: Panel A shows the equilibrium skill distribution in city 1, and panel B shows the distribution in city 2. Workers with skill $s \in [s_2^*, s_1^*)$ can buy a house in city 2 but not in city 1. Some of them—the “missing middle”—choose to locate in city 2 only because they can buy a house there. This increases the s' -low-skilled and the s'' -high-skilled employment shares in city 1.

(b) Figure 3: illustration of Proposition 1.

7.6 Proposition 1 excerpt: the formal mechanism

Read:

- Proposition 1 compares two cities with different price–wage and price–rent ratios.
- The more expensive city has a larger low-skilled share and, under a bounded price–rent condition, a larger high-skilled share.
- Therefore its middle-skilled share is lower.

Economic message: This is the formal statement of the paper’s mechanism: housing-cost differences alone can create city-level employment polarization.

$$n(s, s') \text{ share of } s' \text{ in } s \text{ is given by } n_1^+(s, s') \equiv 1 - n_1^-(s, s') - n_1^*(s, s').$$

To facilitate analysis, consider a simplified version of the model with two cities, 1 and 2, exogenous housing supply $\bar{H}_i$, no individual ownership costs (i.e. $\delta_i = 0$), and no agglomeration externalities (i.e. $\theta = 0$).²⁶ Furthermore, in order to verify that theoretical results do not depend on local differences in skill returns, assume that the returns to skill are the same in both cities, i.e. $a_1(s) = a_2(s)$.

The following proposition constitutes the central theoretical result of the paper. It demonstrates that cities with higher price–wage and price–rent ratios have a higher low-skilled share and, under an additional condition, also have a larger high-skilled share than cities with lower ratios. This implies that such cities have smaller middle-skilled share, i.e. they exhibit greater employment polarization.

Proposition 1 (larger polarization in cities with higher price–wage and price–rent ratios).

Let city 1 have higher price–wage and price–rent ratios, i.e. $p_1/A_1 > p_2/A_2$ and $p_1/r_1 > p_2/r_2$.²⁷ Consider arbitrary skill levels s' and s'' such that s' is below the level required to own a house in city 2 and s'' is above the level required to own in city 1, i.e. $0 < s' < s_2^*$ and $s_1^* < s'' < 1$. Also, let the conditions of Lemma 1 hold. Then city 1 has a larger s' -low-skilled

25. Wages only depend on a function that is strictly increasing in skills and an exogenous productivity term. Hence, local skill and wage distributions are isomorphic and I use these two terms interchangeably.

26. Propositions 1 and 2 derived below will most likely hold with $\delta_i > 0$ and $\theta = 0$ but assuming that those terms are zero significantly simplifies the proofs.

Original excerpt: Proposition 1.

7.7 Table 5 + Figure 4: calibration and benchmark fit

Read:

- Table 5 reports the calibrated parameters for the two-location quantitative economy.
- Large CZs have a higher land share, making housing supply less elastic.
- Large CZs have stronger changes in returns to skill, consistent with SBTC.

- Figure 4 shows the benchmark model’s evolution of middle-skilled shares, wage variance, price–rent ratios, and price–wage ratios.
- The large-small gaps widen in both polarization and inequality, and large CZs become especially expensive relative to wages.

Economic message: The benchmark model is calibrated to reproduce the key spatial trends. It then becomes a laboratory for separating SBTC from housing affordability.

TABLE 5
Model parameters

Panel A: Economy-wide time-invariant parameters			
Parameter			
Fréchet elasticity		$\epsilon = 6.1$	
Agglomeration externality		$\theta = 0.04$	
PTI constraint		$\lambda = 0.308$	
Annual discount factor		$\beta = 0.96$	
Panel B: Local time-invariant parameters			
Parameter	CZ group		
Land share, η_i	Small		0.2239
	Large		0.3965
Panel C: Economy-wide time-varying parameters			
Parameter		1980	2019
Elasticity of utility w.r.t. housing, γ		0.204	0.255
Minimum owner-occupied size, $\bar{h}$		2.68	1.99
Panel D: Local time-varying parameters			
Parameter	CZ group	1980	2019
Productivity, A_{it}	Small	1.000	1.000
	Large	1.088	1.023
Amenities, X_{it}	Small	1.00	1.00
	Large	0.98	1.12
Construction productivity X land, $(\phi_{it}\Lambda_{it})^{\eta_i}$	Small	1.00	1.00
	Large	0.67	0.65
Owners’ expenditures (annualized), δ_{it}	Small	0.032	0.046
	Large	0.039	0.035
Real estate managers’ expenditures (annualized), $\bar{\delta}_{it}$	Small	0.000	0.021
	Large	0.007	0.010
Middle-skilled amenity shifter, x_{it}^M	Small	0.99	1.04
	Large	1.02	0.97
High-skilled amenity shifter, x_{it}^H	Small	0.97	1.05
	Large	1.02	0.94
Skill dispersion, a_{it}	Small	1.81	2.09
	Large	1.84	2.26

Notes: Panel A lists model parameters common to all years and locations. Panel B shows parameters that differ across locations but are constant over time. Panel C shows parameters that vary over time but are common to all locations. Panel D lists parameters that vary by year and location.

(a) Table 5: model parameters.

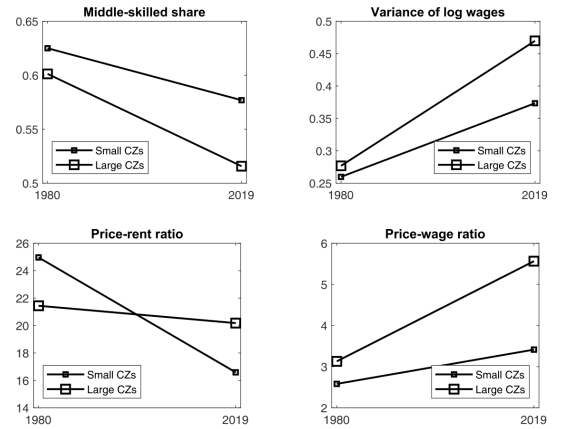


FIGURE 4
Polarization, inequality, and house prices in large and small CZs
Notes: The top two panels plot the middle-skilled share and the variance of log wages in the groups of large and small CZs in the model

(b) Figure 4: benchmark evolution.

7.8 Figure 5 + Figure 6 + Table 6: counterfactuals and welfare

Read:

- Figure 5 Panel A shows that shutting down SBTC reduces the large-small polarization and inequality gaps, but does not eliminate the role of housing.
- Figure 5 Panel B shows that holding price–wage and/or price–rent ratios fixed sharply reduces excess polarization in large cities.
- Figure 6 shows that the largest welfare gains from the price–wage counterfactual accrue to middle-skilled workers.
- Table 6 summarizes aggregate changes: constant price–wage ratios raise homeownership and welfare; constant price–rent ratios can reduce output and do not necessarily increase homeownership.

Economic message: Affordability is not a small side channel. It substantially amplifies the spatial effect of SBTC and changes welfare across the skill distribution.

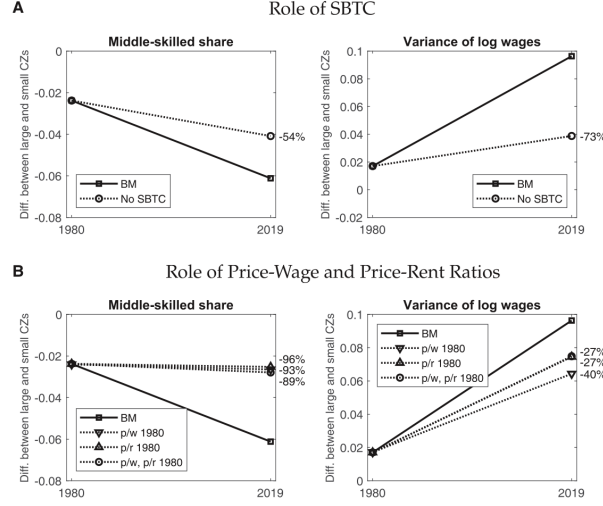


FIGURE 5
Counterfactual results
Notes: The left figure in panel A shows the difference between the share of middle-skilled employment in large CZs and the share in small CZs in the benchmark economy (solid line) and in the counterfactual (dotted line) where SBTC is shut down. The right figure in panel A shows the difference between the variance of log wages in large CZs and the variance in small CZs for the benchmark and the counterfactual economies. Panel B shows results for the counterfactual where the levels of price-wage and price-rent ratios are fixed at their 1980 levels. The numbers to the right of each panel report the percentage change from the benchmark to the counterfactual economy.

Figure 5: counterfactual results.

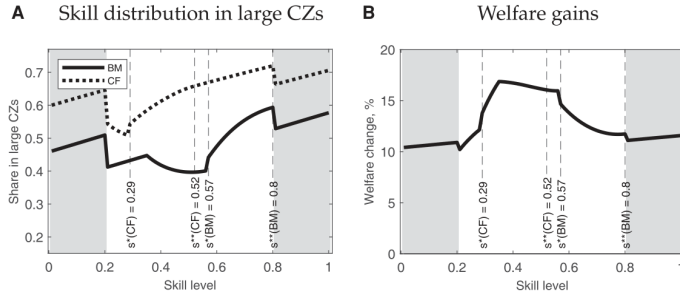


FIGURE 6
Results by skill
Notes: Panel A shows the fraction of workers of each skill level that resides in large CZs in the benchmark economy (solid line) and the counterfactual economy (dashed-dotted line) where price-wage ratios remain constant at the 1980 level. Panel B shows welfare gains in the counterfactual scenario relative to the benchmark for workers at each skill level. In both panels, vertical lines show the skill thresholds for homeownership in benchmark and counterfactual economies.

(a) Figure 6: results by skill.

TABLE 6
Counterfactual changes

	Benchmark	Const. p/w	Const. p/r	Const. $p/w, p/r$
Homeownership rate, %	61.0	78.2	52.3	68.1
Employment in large CZs, %	49.2	64.7	40.3	50.2
Output	100	102.1	98.4	99.9
Welfare	100	112.4	98.6	112.4

Notes: The table shows the values of several variables of interest in the benchmark economy calibrated to 2019 and the counterfactual economy.

(b) Table 6: counterfactual changes.

8 Short summary

- Large U.S. cities experienced more job polarization and greater wage inequality growth from 1980 to 2019.
- Large cities also experienced faster increases in house prices and price-wage ratios.
- Places with faster housing-cost growth lost more middle-skilled employment and saw larger increases in inequality.
- The relationship is visible in OLS regressions and in IV regressions using land unavailability interacted with real interest-rate changes.
- Middle-income households are more likely than low- or high-income households to move to cheaper states for housing-related reasons.
- The model explains this through a tenure threshold: low-skill households cannot buy, high-skill households can buy everywhere, and middle-skill households can buy only where affordability is high enough.

- Expensive cities lose middle-skill households to cheaper cities, creating a “missing middle” and greater local inequality.
- Quantitatively, SBTC remains a major driver of the large-city inequality and polarization gaps.
- But housing affordability amplifies SBTC: keeping price–wage or price–rent ratios at 1980 levels removes most of the excess polarization and a large share of excess inequality.
- The biggest welfare losses from rising large-city housing costs fall on middle-skilled workers.
- Housing supply reforms in large productive cities can improve spatial allocation and reduce local inequality.

9 Conclusion

9.1 Core conclusion

Parkhomenko argues that the housing market is a central force behind the rise of polarization and inequality in large cities. The paper does not claim that housing replaces skill-biased technical change. Instead, housing affordability determines how the labour-market consequences of SBTC are distributed across space. When prices rise faster than wages and rents in large cities, the middle of the skill distribution increasingly cannot access homeownership there and relocates to cheaper places.

9.2 Economic intuition

The core intuition is a threshold mechanism. Low-skill households are renters almost everywhere. High-skill households can own almost everywhere. Middle-skill households are on the margin of ownership. Therefore, increases in price–wage and price–rent ratios affect the middle most strongly. As large cities become expensive, the middle-skilled move out, while low-skilled renters and high-skilled owners remain. This creates a polarized local skill distribution and a more dispersed wage distribution.

9.3 Takeaway

The paper links urban housing affordability to labour-market inequality. It shows that housing supply constraints and homeownership incentives can transform productivity growth into local polarization. For policy, the most important lesson is that making productive cities affordable is not only a housing issue; it is also a labour-market inequality issue.